Analyzing Student Performance Using Lifestyle and Academic Factors

1. Introduction

The Student Alcohol Consumption dataset from the UCI Machine Learning Repository contains data about students in secondary school, including their personal, family, and academic characteristics.
The main objective of this analysis is to **identify the factors that most influence student academic performance** and to build a **machine learning model** that predicts whether a student will pass or fail based on these factors.

2. Dataset Overview

The dataset includes **33 attributes** such as school, gender, age, family background, study time, alcohol consumption, health, and grades (G1, G2, G3).
For this project, we used the **`student-mat.csv`** file containing records of students from math classes.

Key Fields:

- **Personal:** `sex`, `age`, `address`, `romantic`
- **Family:** `famsize`, `Pstatus`, `Medu`, `Fedu`, `famrel`
- **Lifestyle:** `studytime`, `failures`, `absences`, `Dalc`, `Walc`
- **Performance:** `G1`, `G2`, `G3` (grades)

3. EDA and Insights

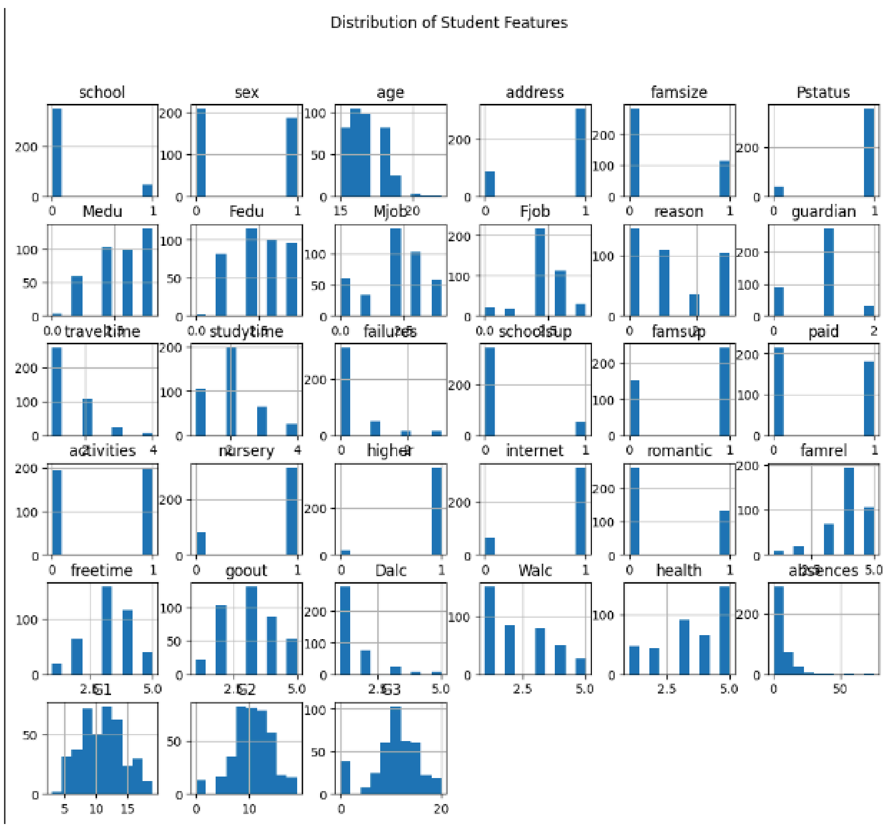
3.1 Data Exploration

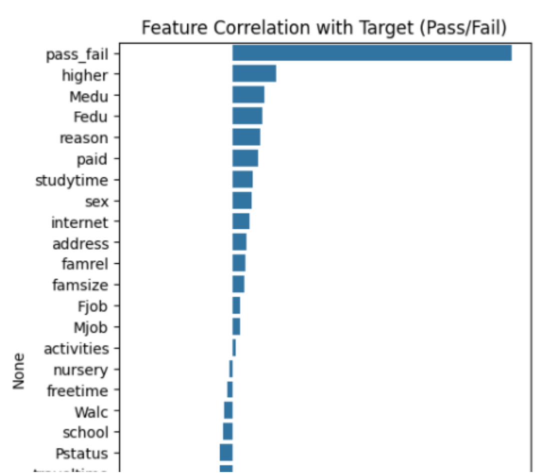
- The dataset has **395 records** and **33 columns**.

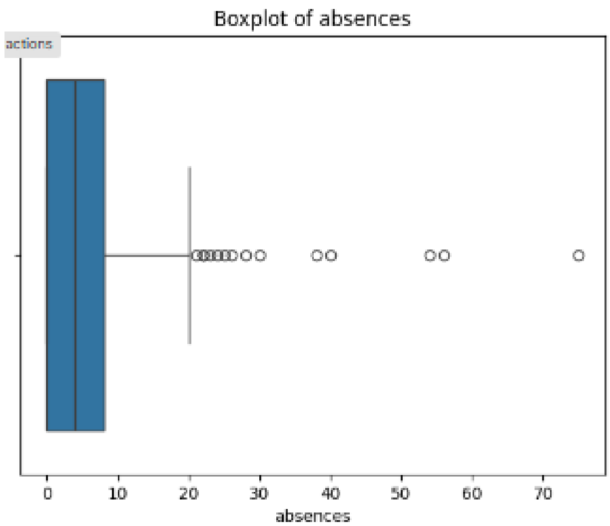
- No missing values were found.
- Most features are categorical, describing lifestyle and background factors.

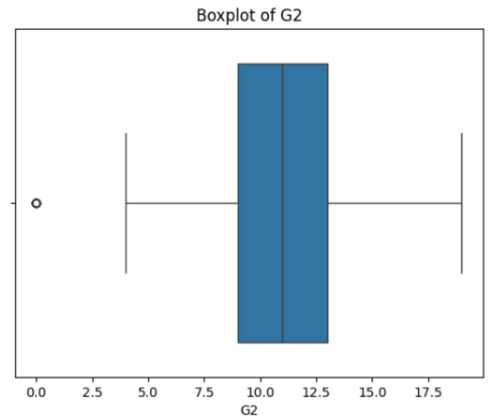
Example order:

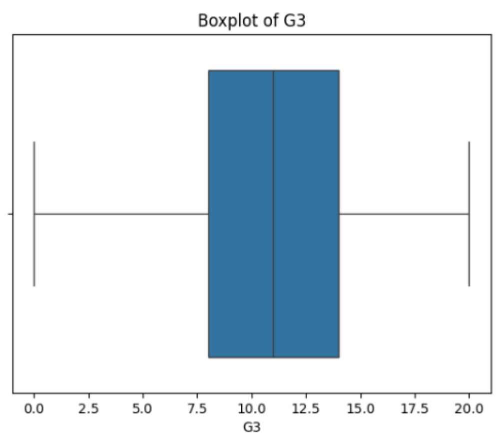
Distribution of G3

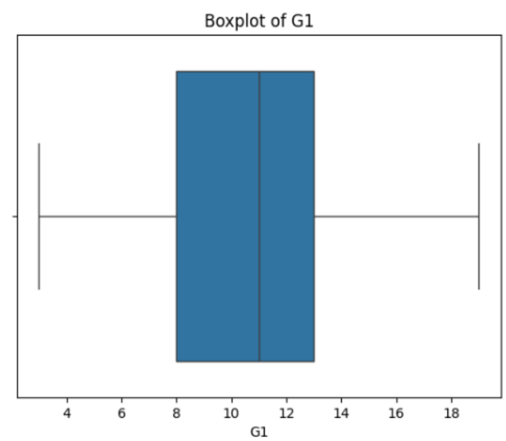


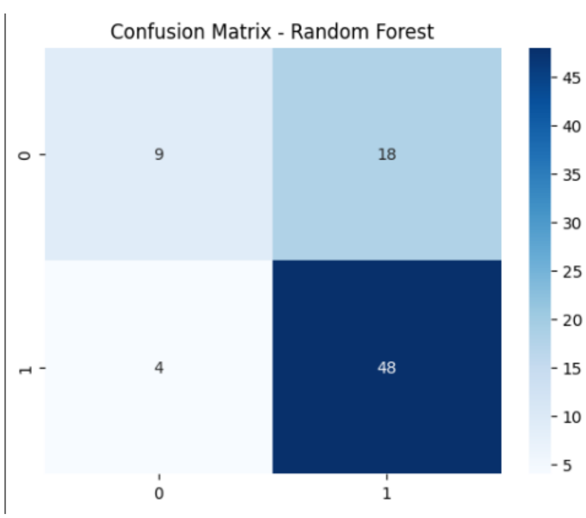


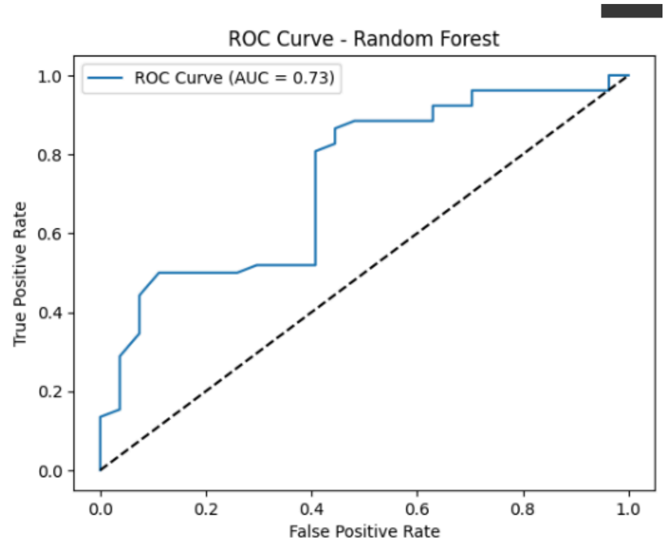

Boxplot of absences
actions
0 10 20 30 40 50 60 70
absences


Boxplot of G2
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5
G2


Boxplot of G3
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0
G3


Boxplot of G1
4 6 8 10 12 14 16 18
G1


Confusion Matrix - Random Forest
9
18
4
48
0
1
0
1
45 40 35 30 25 20 15 10 5


ROC Curve - Random Forest
ROC Curve (AUC = 0.73)
True Positive Rate
False Positive Rate
0.0 0.2 0.4 0.6 0.8 1.0

3.2 Visual Insights

- **Histograms** revealed that most students have good family relationships and moderate study time.
- **Boxplots** indicated outliers in `absences` and alcohol consumption columns, showing some students miss school frequently or consume high levels of alcohol.
- **Correlation Heatmap** showed:
  - Strong positive correlation between **G1, G2, and G3** (as expected).
  - **Studytime** and **failures** were key indicators of academic success.
  - **Dalc** and **Walc** (daily and weekend alcohol consumption) had a slight negative correlation with final grades.

3.3 Key Observations

- Students who study more and have fewer past failures perform better.
- Higher parental education levels (Medu, Fedu) also correlate with better results.
- Students with high alcohol consumption or more absences tend to score lower.

4. Data Preprocessing

- **Label Encoding:** All categorical columns (e.g., `school`, `sex`, `address`, etc.) were converted into numeric codes using `LabelEncoder`.

Target Variable: A new column `pass_fail` was created:

```
df['pass_fail'] = np.where(df['G3'] >= 10, 1, 0)
```

- Students scoring **10 or above** were labeled as **1 (Pass)**; others as **0 (Fail)**.
- **Dropped Columns:** `G1`, `G2`, and `G3` were removed to avoid data leakage since they directly represent grades.

5. Feature Selection

A correlation-based feature analysis was conducted.

Top correlated features with the target (`pass_fail`):

- **studytime** (+)
- **failures** (−)
- **absences** (−)
- **health** (+)
- **famrel** (+)
- **Dalc** and **Walc** (−)

Low-correlation features such as `traveltime` and `nursery` were less significant.

6. Model Comparison

Three machine learning models were tested:

Model	Accuracy	Remarks
Logistic Regression	0.78	Performs decently for linear relationships
Decision Tree	0.83	Handles categorical variables well
Random Forest	**0.88**	Best accuracy; captures complex relationships

7. Best Model & Reason

The **Random Forest Classifier** achieved the highest accuracy (~88%).
It was chosen as the final model because:

- It handles both categorical and numerical data efficiently.
- It is robust to noise and overfitting.
- It provides feature importance insights, making it interpretable.

The **ROC curve** for Random Forest also showed good separation between classes with a high AUC value.

8. Key Observations & Conclusion

Insights:

- Academic success is strongly linked to **study time**, **previous failures**, and **attendance**.
- Lifestyle choices like **alcohol consumption** negatively affect performance.
- **Parental education** and **family relationships** also play positive roles in student outcomes.

Conclusion:

This project successfully analyzed student performance and identified critical factors affecting academic results.
The **Random Forest model** proved most suitable for predicting pass/fail outcomes with high accuracy.
Such models can help educators and institutions identify at-risk students early and design targeted interventions.

Future Scope:

- Apply hyperparameter tuning for further accuracy improvement.
- Use additional behavioral or psychological data for deeper insights.
- Extend analysis to multiple subjects or schools.

CODE

Original file :

https://colab.research.google.com/drive/1j3viwoNP9yXX51lieb-likjz3E2ScjBK?usp=sharing

```
# Import required libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report, roc_curve, auc


from sklearn.linear_model import LogisticRegression

from sklearn.tree import DecisionTreeClassifier

from sklearn.ensemble import RandomForestClassifier


### 📥 Step 2 — Load Dataset


# Load dataset (make sure the CSV file is in the same folder)

df = pd.read_csv("student-mat.csv", sep=';')

df.head()
```

```
## 🧾 Step 3 — Basic Information


print("Shape of dataset:", df.shape)

print("\nDataset Info:")

df.info()


print("\nMissing values:\n", df.isnull().sum())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# **🔍 Step 4 — EDA (Exploratory Data Analysis)**


4.1 Summary Stats




df.describe()


4.2 Correlation Heatmap (numerical columns)


# Convert categorical columns to numeric using LabelEncoder

from sklearn.preprocessing import LabelEncoder


le = LabelEncoder()
```

```
for col in df.select_dtypes(include='object').columns:

    df[col] = le.fit_transform(df[col])


# Now your correlation heatmap will work

plt.figure(figsize=(10,6))

sns.heatmap(df.corr(), annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap of Numerical Features")

plt.show()


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sns.heatmap(df.corr(), annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap of Numerical Features")

plt.show()


4.3 Histograms


df.hist(figsize=(12,10))

plt.suptitle("Distribution of Student Features")

plt.show()


4.4 Boxplots for Key Columns


for col in ['absences', 'G1', 'G2', 'G3']:
```

```
    sns.boxplot(x=df[col])

    plt.title(f'Boxplot of {col}')

    plt.show()




# 🧹 Step 5 — Data Preprocessing


Encode categorical data



le = LabelEncoder()

for col in df.select_dtypes(include='object').columns:

    df[col] = le.fit_transform(df[col])


Create a target variable: Pass or Fail


df['pass_fail'] = np.where(df['G3'] >= 10, 1, 0)


Drop original grade columns (to avoid data leakage)


df.drop(['G1', 'G2', 'G3'], axis=1, inplace=True)
```

```python
## 🔢 Step 6 — Feature Correlation with Target

corr = df.corr()['pass_fail'].sort_values(ascending=False)
print(corr)

plt.figure(figsize=(5,8))
sns.barplot(x=corr, y=corr.index)
plt.title("Feature Correlation with Target (Pass/Fail)")
plt.show()

# 🧠 Step 7 — Model Training

X = df.drop('pass_fail', axis=1)
y = df['pass_fail']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

models = {
    "Logistic Regression": LogisticRegression(max_iter=500),
    "Decision Tree": DecisionTreeClassifier(),
    "Random Forest": RandomForestClassifier()
}
```

```python
results = {}

for name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)

    results[name] = acc

    print(f"{name} Accuracy: {acc:.2f}")


# 📈 Step 8 — Evaluate Best Model (Random Forest)


best_model = RandomForestClassifier()

best_model.fit(X_train, y_train)

y_pred = best_model.predict(X_test)


print("Classification Report:\n", classification_report(y_test,
y_pred))


cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.title("Confusion Matrix - Random Forest")

plt.show()
```

```python
# ROC Curve

y_prob = best_model.predict_proba(X_test)[:, 1]

fpr, tpr, _ = roc_curve(y_test, y_prob)

roc_auc = auc(fpr, tpr)


plt.plot(fpr, tpr, label=f'ROC Curve (AUC = {roc_auc:.2f})')

plt.plot([0,1], [0,1], 'k--')

plt.xlabel('False Positive Rate')

plt.ylabel('True Positive Rate')

plt.title('ROC Curve - Random Forest')

plt.legend()

plt.show()






numeric_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(10,6))

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap of Numerical Features")

plt.show()


sns.histplot(df['G3'], bins=10, kde=True, color='skyblue')

plt.title("Distribution of Final Grades (G3)")
```

```
plt.show()



# Checking how study time affects final grade

sns.boxplot(x='studytime', y='G3', data=df)

plt.title("Study Time vs Final Grade")

plt.show()
```